

Ahmad Najmuddin Ibrahim

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Education

Ibaraki University , Industrial Science (Robotics Engineering)	Hitachi, Japan
- Research on assistive grouser mechanisms for wheeled rover locomotion on soft sand inclines. - Developed prototype rovers tested on unconsolidated sandy terrain.	2013 – 2017
Ibaraki University , Intelligent Systems Engineering	Hitachi, Japan
Research on prototype adaptable feet sole mechanism using vacuum-jamming mechanism	2011 – 2013
Ibaraki University , Intelligent Systems Engineering	Hitachi, Japan
	2008 – 2011

Experience

Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA) , Senior Lecturer (Pensyarah Universiti, DS13-A)	Pekan, Pahang, Malaysia
- Teaching undergraduate courses in robotics, mechatronics, and control systems. - Supervising PhD and Master's students in robotics and mechatronics research. - Principal Investigator for HVAC duct monitoring robot research project.	Dec 2024 – present 1 year 8 months
Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA) , Pensyarah Universiti (DS51-A)	Pekan, Pahang, Malaysia
- Taught core courses including Microcontroller System, Robotic Prototype Design, and Robotics System Modelling. - Served as Head of Programme for Bachelor of Mechatronic Engineering Technology (Robotics).	Feb 2021 – Nov 2024 3 years 10 months
Universiti Malaysia Pahang , Lecturer	Pekan, Pahang, Malaysia
- Developed and delivered robotics and mechatronics courses. - Initiated research programme on assistive grouser mechanisms for wheeled rovers.	May 2017 – Jan 2021 3 years 9 months

Awards

GOLD Medal — ITEX 2024: International Invention, Innovation & Technology Exhibition (ITEX) — Awarded for the project 'Remotely Underwater Cleaning Robot'.
SILVER Medal — ITEX 2023: International Invention, Innovation & Technology Exhibition (ITEX) — Awarded for the project 'Agronetics IoT Based Smart Farming System'.

Skills

Robotic Mechanical System Design

Mechanical-Electronics Systems Integration

Discrete Element Modelling (DEM)

Mobile Robotics

Interests

Terrain-Adaptive Robot Mobility: Grouser mechanisms, soft-soil terramechanics, wheeled rover locomotion, traction optimisation

Field & Inspection Robotics: HVAC duct inspection, confined-space navigation, tethered robotic systems

Agricultural Robotics: Autonomous field robots, pest deterrence, unstructured terrain navigation

Embedded Systems & Control: Microcontroller systems, PID control, sensor fusion, real-time embedded programming

Languages

Malay

Native speaker

English

Fluent

Japanese

Conversational

Projects

Automated Cleaning and Monitoring of HVAC Duct Using Mobile Robot 2024 – 2026

Development of a tethered mobile robot platform for automated HVAC duct inspection and cleaning, integrating visual and environmental sensors for condition assessment.

- Grant reference RDU243812, UIC240849
- Principal Investigator

Grouser Effect Contact Modelling in Cohesive Unstructured Terrains 2023 – 2025

Investigation of grouser-soil interaction mechanisms in cohesive unstructured terrains using experimental and simulation approaches.

- Grant reference RDU230135
- Principal Investigator

Development of Autonomous Mobile Robot for Optimum HVAC Duct Cleaning 2021 – 2022

Design and development of an autonomous mobile robot for HVAC duct cleaning applications.

- Grant reference RDU212402, UIC210810
- Principal Investigator

Certificates

Graduate Engineer (Manufacturing) Jan 2018

IEEE International Professional Member Jan 2018

Graduate Technologist Jan 2018